

Video Grounded Reverse SFT

Real-World Videos

~1.05M clips,
10 categories



...

Select v_i

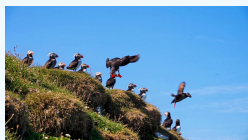
Captioner

Cinematic Condition y_i

Video-level Overview: ... This nature documentary clip observes a puffin colony taking flight on a coastal cliff.... The visual style is bright and , ...employs a stable, observational perspective...telephoto lens...accompanied by a gentle orchestral score ...

Shot-level Plan

shot1 [00:00-06:36]



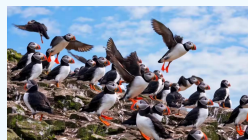
a static time-lapse long shot...
hard sunny daytime light ... A
puffin dives down rapidly from ...
a gentle string melody begins...

shot2 [06:36-12:08]



... a static close-up medium...
four puffins... slowly turn their
heads...gentle orchestral music
continues... throughout the shot...

shot3 [12:08-19:32]



... a static eye-level medium shot...
...larger congregation of puffins...
lit by hard sunny light... take off
and fly away upwards ... together...

shot4 [19:32-26:20]



... a static eye-level long shot...
soft overcast light...puffins
across the sky, in flocks ...gentle
orchestral music fading out ...

2K human
requests

few-shot

LLM User-Request Reconstruction

x_i

SFT Training Pair

User Request x_i + Cinematic Condition y_i

Prompt Enhancer
 π_{SFT}

Semantic Consistency GRPO

SC-GRPO Prompt Construction

Human-written Requests

Broad-coverage
Human Requests

Challenging Cases

π_{SFT}

wan3.0's video generator

Human inspection

SC-GRPO Dataset (~15K): Broad Coverage + Challenging Cases

Prompt Enhancer π_θ

$y^{(1)}$

$y^{(2)}$

...

$y^{(g)}$

Semantic-Consistency Reward $r_{sem}(x, y^{(g)})$

Style

Subjects

Action

Dialogue

Sound

Camera

Lighting

Spatial Relations

Scene

Cross-shot consistency is assessed within every dimension

Group-relative Advantages $\hat{A}_g$

SC-GRPO Policy Update